

EDUCATION

- **(Top 50) Monash University** Melbourne, Australia
PhD Degree (Supervised by Prof. Aldeida Aleti, Prof. Chunyang Chen, Prof. Hongyu Zhang) Oct 2022 – Apr 2026
 ◦ Visiting Scholar: NLP Lab @ THU (Oct 2024 – Jun 2025), SE&AI Lab @ TUM (Jun 2025 – Aug 2025)
- **(Top 100) KTH Royal Institute of Technology** Stockholm, Sweden
Master's Degree in Machine Learning (Thesis Supervisor: Prof. Martin Monperrus) Aug 2018 – Aug 2020
- **(Project 985) Shandong University (the Elite Program)** Jinan, China
Bachelor's Degree in Computer Science (Thesis Supervisor: Prof. Jun Ma) Sep 2012 – Jun 2016

SELECTED PAPERS

- **Distilling Language Models with Capacity-Aware Reaction Transfer**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang, Peng Di, Sheng Gao Under Review
- **SemRF: A Semantic Reference Frame for Residual-Stream Dynamics in LMs**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang Under Review
- **Rethinking Weight Tying: Pseudo-Inverse Tying for LM Stable Training and Updates**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang Under Review
- **Cross-Scale Knowledge Transfer in LMs through Latent Semantic Alignment** ACL'26
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (CORE A*)
- **Semantic-based Optimization for Repairing LLMs: Case Study on Code Generation** ICSE'26
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (CORE A*)
- **Semantic-Aware Layer-Freezing for Computation-Efficient Fine-Tuning of LMs** ACL'25
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (CORE A*)
- **Vocabulary-Defined Semantics: Latent Space Clustering for Beyond-Context Learning**
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang Under Review
- **Neuron Patching: Semantic-based Neuron-level LM Repair for Code Generation** TOSEM 2026
Jian Gu, Aldeida Aleti, Chunyang Chen, Hongyu Zhang (JCR Q1)
- **Towards Top-Down Automated Development in Limited Scopes: A Neuro-Symbolic Framework from Expressibles to Executables** FSE'23 (IVR)
Jian Gu, Harald C. Gall (CORE A*)
- **Assemble Foundation Models for Automatic Code Summarization** SANER'22
Jian Gu, Pasquale Salza, Harald C. Gall (CORE A)
- **Multimodal Representation for Neural Code Search** ICSME'21
Jian Gu, Zimin Chen, Martin Monperrus (CORE A)
- **Source-Free Detection and Impact Analysis of Compiler Optimization in Mobile Apps** ASE 2026
Han Hu, Xiaoheng Xie, Bo Sun, Jian Gu, Gang Fan, Li Li (CORE A*)
- **On the Effectiveness of Transfer Learning for Code Search** TSE 2022
Pasquale Salza, Christoph Schwizer, Jian Gu, Harald C. Gall (JCR Q1)
- **Automated Classification of Overfitting Patches with Statical Code Features** TSE 2021
He Ye, Jian Gu, Matias Martinez, Thomas Durieux, Martin Monperrus (JCR Q1)

WORK EXPERIENCE

- **Ant Group** Hang Zhou, China
Research Intern Feb 2026 – Aug 2026
 - **LM Distillation**: proposed a novel distillation technique for LMs, enabling better support for edge-device tasks.
- **Huawei Research Center** Hong Kong, China
Research Intern Sep 2025 – Jan 2026
 - **Byte-Level Encoder**: compiler provenance recovery for performance load reduction in HarmonyOS ecosystem.
 - **On-Device Small LMs**: Small LM pretraining and post-training for intelligent coding in HarmonyOS IDE.
- **Monash University** Melbourne, Australia
Teaching Assistant and Co-Supervisor Jul 2023 – Jun 2024
 - **Teaching**: FIT5003 Software Security (1 semester); FIT4701/FIT4702 Projects and Research Theses (1 year).
- **University of Zurich** Zurich, Switzerland
Research Assistant, PhD Candidate Sep 2020 – Sep 2022
 - **Research**: Conducted research on AI for SE, including code retrieval, comment generation, and code generation.
 - **Teaching**: Designed and advised “AI Pair Programming”, a Master’s team project on code intelligence (1 year).
- **KTH Royal Institute of Technology** Stockholm, Sweden
Research Engineer Jun 2019 – Aug 2019
 - **Eclipse Repairnator**: Repairnator is a pioneering bot for automated program repair, hosted by the eclipse foundation. Integrated [Coming](#) and [SequenceR](#) into it and implemented its official GitHub app. Implemented Prophet4J (incl. Sketch4Repair) to generate code patches. Implemented Context2Name to prettify test cases.
- **Smart Software Studio** Yancheng, China
Software Engineer Jul 2017 – Jun 2018
 - **YoYo**: Delivered a technical solution including websites and apps for online community, with the supports for local advertisements. It is for identity-based socials, mainly communications between alumni and neighbors.
 - **WiiPM**: Developed a 2D rogue-like game on the mobile platform. WiiPM contains elements of turn-based battles as well as collection and role-plays adventure. The technique of procedural content generation is adopted.
- **UCar Inc.** Beijing, China
Software Engineer Mar 2016 – Nov 2016
 - **Anomaly Detection for Anti-Fraud**: Analyzed unusual operational logs, summarized anomaly patterns.
 - **Streaming Data Pipeline and Panel**: Maintained the streaming data pipeline for real-time transactions.
- **Hisense Group (R&D Center)** Qingdao, China
Algorithm Intern Sep 2015 – Feb 2016
 - **Movie GenePool**: Improved online recommender systems (collaborative filtering). Extracted plot tags and review tags to build movie knowledge base. Designed unified knowledge base to fuse multiple data providers. Developed one workflow system collecting and processing movie metadata as well as one demonstrative recommender system.

SERVICES

- **Software Engineering (IEEE TSE, ACM TOSEM)** Reviewer
- **Machine Learning (ICLR, NIPS, AAI)** PC Member, Reviewer
- **Multidisciplinary (WWW, IEEE TAI)** PC Member, Reviewer

HONORS & GRANTS

- **Government Award for Outstanding Students Abroad** Chinese Government 2025 – 2026
- **Top-Up Scholarship & Supplementary Funding** Monash University 2025 – 2026
- **\$80K (2000 kSU) Research Grant** AU National Computational Infrastructure 2024 – 2026
- **\$30K HPC Research Grant** DUG Technology Ltd 2023 – 2024
- **Ranked #1 (until Jun 2020)** GitHub CodeSearchNet Challenge 2019 – 2020